Lab Experiment 10

Introduction to Engineering Technology Programming

Learning Outcomes

By the end of these activities, students will be able to:

1. **Interpret an engineering equation** by identifying variables, units, and relationships in the cantilever beam deflection formula.
2. **Write MATLAB and Python programs** that prompt the user to input values, perform calculations, and display results with proper units.
3. **Create reusable functions** in both languages to compute beam deflection for different input sets.
4. **Validate and compare** outputs between MATLAB and Python to ensure accuracy and consistency.
5. **Visualize results** by generating a plot of deflection versus beam length using built-in plotting libraries.

Lab Exercise

Objective:

To introduce students to basic programming concepts for engineering calculations using MATLAB and Python. Students will write scripts and functions that calculate the deflection of a cantilever beam subjected to a point load at the free end. They will collect user input for all variables, compute the deflection, and produce a graph showing how beam length affects deflection.

Instructions

A cantilever beam is a structural element that is fixed at one end and free at the other. This configuration allows the beam to carry loads without additional vertical support at the free end. Cantilever designs are common in bridges, balconies, overhanging roofs, traffic-signal supports, and aircraft wings, where clear spans are needed and supports are only available on one side.

The vertical deflection δ at the free end of a cantilever subjected to a point load can be predicted using engineering mechanics. In this lab, students will implement the classic deflection equation:

$$\delta = \frac{FL^3}{3EI}$$

Where:

- F: Applied load (Newtons)
- L: Beam length (meters)
- E: Young's modulus of the beam material (Pascals)
- I: Second moment of area (m^4)

Understanding this equation is essential for predicting how materials deform under load and for ensuring the safety and serviceability of structural components.

To express this formula computationally, we will use MATLAB and Python. Both are widely used in engineering for numerical analysis, data visualization, and automation. MATLAB offers a math-oriented environment, which is ideal for quick calculations and plotting, while Python is a versatile open-source programming language popular in engineering, scientific computing, and automation tasks.

Part A. Python Implementation

1. Open a Python IDE or notebook and create a new file beam_deflection.py.
2. Write a function beam_deflection(F, L, E, I) that returns delta.
3. Use input() to collect user values and call the function.
4. Print the result formatted to three decimal places in millimeters.
5. Use **NumPy** and **Matplotlib** to generate and display a plot of deflection vs. length for a range of L values.

Part B. MATLAB Implementation

1. Create a **function file** deflection_calc.m that accepts F, L, E, I as inputs and returns delta.
2. Open MATLAB and create a new **script** named beam_deflection.m.
3. Use the input() function to prompt the user for F, L, E, and I.
4. Calculate the deflection delta in meters and convert to millimeters.
5. Print the result using fprintf with proper formatting.
6. Use vector operations to compute deflection for a range of lengths and plot **delta vs. L**.

Part C. Verification and Reflection

1. Run both the MATLAB and Python programs with the same input values and confirm that the calculated deflections match.
2. Prepare a brief written comparison of your experience coding in the two languages. Discuss syntax differences, plotting features, and ease of use.
3. Which programming language did you personally find easier to learn and debug, and why?
4. If you were asked to extend the work done on this lab (For example, adding multiple load cases or automating a report), which language do you think would make development easier.
5. Describe at least one strength of MATLAB and one strength of Python that you would keep in mind when selecting a tool for future projects.

Deliverables:

Submit the following files for both programming languages so that I can verify your work:

- Your Python script, **beam_deflection.py**
- Your MATLAB script, **beam_deflection.m**
- Your MATLAB function file, **deflection_calc.m**
- **A screenshot of the plots** created by your MATLAB and Python scripts (2 in total)

Grading Rubric

- Part A - Python script: Script completeness (0-6 points)
- Part B - MATLAB Script: Script + Function completeness (0-8 points)
- Part C: Completeness of each question, 5 in total. (5*1.5 =7.5 points max)
- Total Points: 20 + 1.5 (extra)